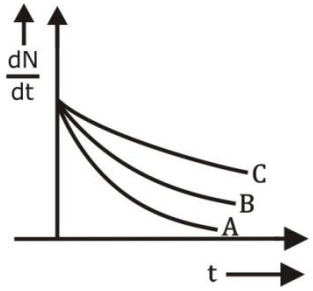
KCET-2019 (Physics)

1. Which one of the following nuclei has shorter mean life?

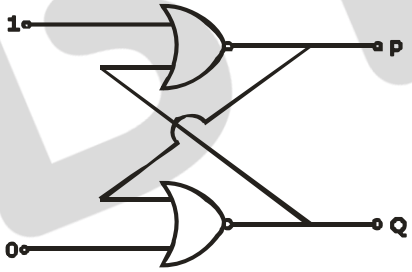


   a. B
   b. Same for all
   c. A
   d. C

2. The conductivity of semiconductor increases with increase in temperature because
   a. Relaxation time increases
   b. Number density of current carriers increases, relaxation time decreases but effect of decrease in relaxation time is much less than increase in number density
   c. Number density of charge carriers increases
   d. Both number density of charge carriers and relaxation time increase

3. For a transistor amplifier, the voltage gain
   a. Is high at high and low frequencies and constant in the middle frequency range
   b. Constant at high frequencies and low at low frequencies
   c. Remains constant for all frequencies
   d. Is low at high and low frequencies and constant at mid frequencies

4. In the following circuit, what are P and Q?



   a. P = 1, Q = 0
   b. P = 1, Q =1
   c. P = 0, Q = 0
   d. P = 0, Q =1

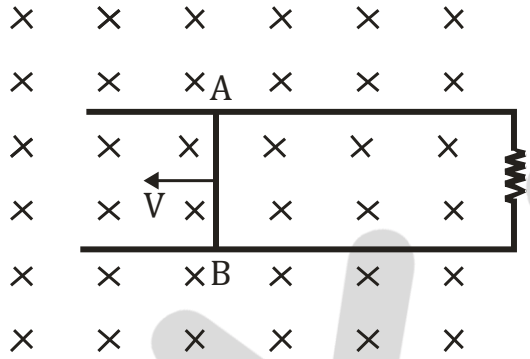
5. An antenna uses electromagnetic waves of frequency 5 MHz. For proper working, the size of the antenna should be
   a. 300 m
   b. 3 km
   c. 15 m
   d. 15 km

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6. A magnetic needle has a magnetic moment of 5 x 10^{-2} Am^2 and moment of inertia 8 x 10^{-6} kgm^2. It has a period of oscillation of 2s in a magnetic field $\vec{B}$. The magnitude of magnetic field is approximately.
   a. 0.4×10^{-4}T
   b. 0.8×10^{-4}T
   c. 1.6×10^{-4}T
   d. 3.2×10^{-4}T

7. A toroid has 500 turns per metre length. If it carries a current of 2A, the magnetic energy density inside the toroid is
   a. 0.314 J/m^3
   b. 3.14 J/m^3
   c. 0.628 J/m^3
   d. 6.28 J/m^3

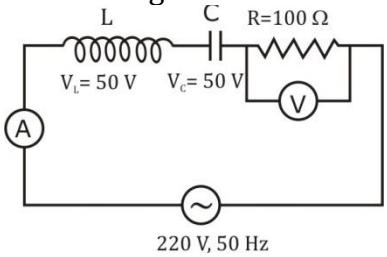
8. Consider the situation given in figure. The wire AB is slid on the fixed rails with a constant velocity. If the wire AB is replaced by a semicircular wire, the magnitude of the induced current will



   a. Remain same
   b. Increase or decrease depending on whether the semicircle bulges towards the resistance or away from it
   c. Increase
   d. Decrease

9. The frequency of an alternating current is 50 Hz. What is the minimum time taken by current to reach its peak value from rms value?
   a. 2.5×10^{-3}s
   b. 10×10^{-3}s
   c. 5×10^{-3}s
   d. 0.02s

10. The readings of ammeter and voltmeter in the following circuit are respectively



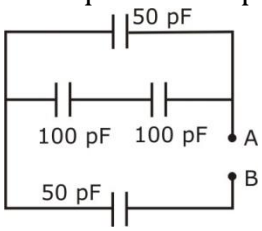
   a. 1.5 A, 100 V
   b. 2.2 A, 220 V
   c. 1.2 A, 120 V
   d. 2.7 A, 220 V

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11. Two metal plates are separated by 2 cm. The potentials of the plates are -10 V and + 30 V. The electric field between the two plates is
    a. 1000 V/m
    b. 3000 V/m
    c. 500 V/m
    d. 2000 V/m

12. The equivalent capacitance between A and B is,



    a. $\frac{100}{3}$ pF
    b. 300 pF
    c. 50 pF
    d. 150 pF

13. A capacitor of capacitance C charged by an amount Q is connected in parallel with an uncharged capacitor of capacitance 2C. The final charges on the capacitors are
    a. $\frac{Q}{4}, \frac{3Q}{4}$
    b. $\frac{Q}{5}, \frac{4Q}{5}$
    c. $\frac{Q}{2}, \frac{Q}{2}$
    d. $\frac{Q}{3}, \frac{2Q}{3}$

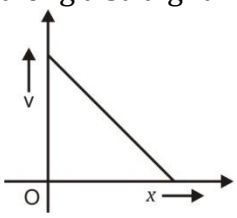
14. Though the electron drift velocity is small and electron charge is very small, a conductor can carry an appreciably large current because
    a. Drift velocity of electron is very large
    b. Relaxation time is small
    c. Electron number density is very large
    d. Electron number density depends on temperature

15. Masses of three wires of copper are in the ratio 1:3:5 and their lengths are in the ratio 5:3:1. The ratio of their electrical resistance are
    a. 5:3:1
    b. 125:15:1
    c. 1:3:5
    d. 1:15:125

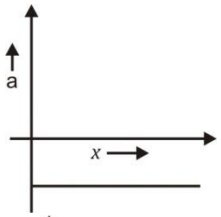
16. If P, Q and R are physical quantities having different dimensions, which of the following combinations can never be a meaningful quantity?
    a. PQ - R
    b. $\frac{PR - Q^2}{R}$
    c. $\frac{P - Q}{R}$
    d. $\frac{PQ}{R}$

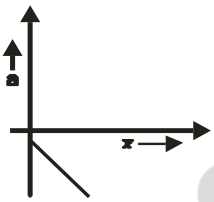
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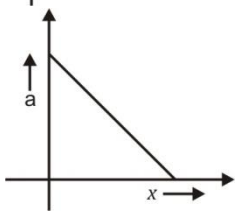
17. The given graph shows the variation of velocity (v) with position (x) for a particle moving along a straight line

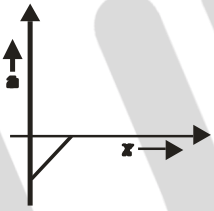


Which of the following graph shows the variation of acceleration (a) with position (x)?

a. 

b. 

c. 

d. 

18. The trajectory of a projectile projected from origin is given by the equation $y = x - \frac{2x^2}{5}$. The initial velocity of the projectile is?
    a. 5 ms^{-1}
    b. $\frac{5}{2} ms^{-1}$
    c. $\frac{2}{5} ms^{-1}$
    d. 25 ms^{-1}

19. An object with mass 5 kg is acted upon by a force, $\vec{F} = (-3\hat{\imath} + 4\hat{\jmath})$ N. If it's initial velocity at $t = 0\ is\ \vec{v} = (6\hat{\imath} - 12\hat{\jmath})$ ms^{-1}, the time at which it will just have a velocity along y-axis is
    a. 10 s
    b. 15 s
    c. 5 s
    d. 2 s

20. During inelastic collision between two objects, which of the following quantity always remains conserved?
    a. Total mechanical energy
    b. Speed of each body
    c. Total kinetic energy
    d. Total linear momentum

21. In Rutherford experiment, for head-on collision of α-particles with a gold nucleus, the impact parameter is
    a. of the order of 10^{-14} m
    b. of the order of 10^{-6} m
    c. zero
    d. of the order of 10^{-10} m

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22. Frequency of revolution of an electron revolving in n^{th} orbit of H-atom is proportional to
    a. n
    b. $\frac{1}{n^3}$
    c. $\frac{1}{n^2}$
    d. n independent of n

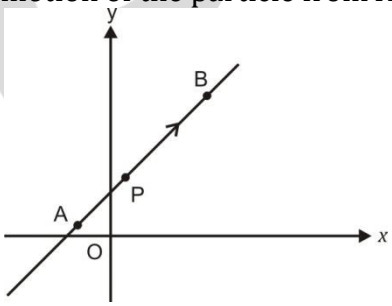
23. A hydrogen atom in ground state absorbs 10.2 eV of energy. The orbital angular momentum of the electron is increased by
    a. 2.11×10^{-34} Js
    b. 4.22×10^{-34} Js
    c. 1.05×10^{-34} Js
    d. 3.16×10^{-34} Js

24. The end product of decay of $_{90}Th^{232}$ is $_{82}Pb^{208}$. The number of α and β particles emitted are respectively.
    a. 6, 4
    b. 4, 6
    c. 3, 3
    d. 6, 0

25. Two protons are kept at a separation of 10 nm. Let F_n and F_e be the nuclear force and the electromagnetic force between them
    a. $F_e >> F_n$
    b. F_e and F_n differ only slightly
    c. $F_e = F_n$
    d. $F_e << F_n$

26. Two particles which are initially at rest move towards each other under the action of their mutual attraction. If their speeds are v and 2v at any instant, then the speed of center of mass of the system is,
    a. Zero
    b. v
    c. 2v
    d. 1.5 v

27. A particle is moving uniformly along a straight line as shown in the figure. During the motion of the particle from A to B, the angular momentum of the particle about 'O'



    a. decreases
    b. first increases then decreases
    c. increases
    d. Remains constant

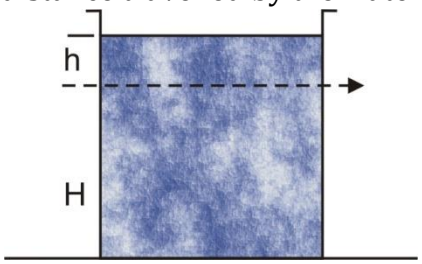
28. A satellite is orbiting close to the earth and has a kinetic energy K. The minimum extra kinetic energy required by it to just overcome the gravitation pull of the earth is
    a. 2K
    b. $2\sqrt{2}K$
    c. K
    d. $\sqrt{3}K$

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29. A wire is stretched such that it's volume remains constant. The Poisson's ratio of the material of the wire is

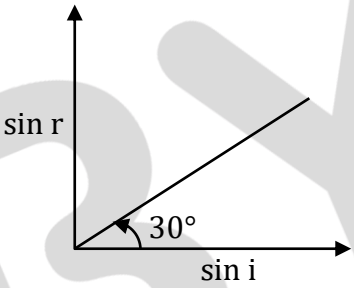
a. –0.50
b. –0.25
c. 0.50
d. 0.25

30. A cylindrical container containing water has a small hole at height of H = 8 cm from the bottom and at a depth of 2 cm from the top surface of the liquid. The maximum horizontal distance travelled by the water before it hits the ground (x) is



a. $4\sqrt{2}$ cm
b. 6 cm
c. 8 cm
d. 4 cm

31. A transparent medium shows relation between i and r as shown. If the speed of light in vaccum is c, the Brewster angle for the medium is



a. 45°
b. 90°
c. 30°
d. 60°

32. In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where path difference is λ is K units. The intensity of light at a point where path difference is $\frac{\lambda}{3}$ is

a. $\frac{K}{4}$
b. 2K
c. K
d. 4K

33. Due to Doppler's effect, the shift in wavelength observed is 0.1 Å for a star producing wavelength 6000 Å. Velocity of recession of the star will be

a. 10 km/s
b. 20 km/s
c. 25 km/s
d. 5 km/s

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34. An electron is moving with an initial velocity $\vec{V} = V_0 \hat{i}$ and is in a uniform magnetic field $\vec{B} = B_0 \hat{j}$. Then, it's de Broglie wavelength
    a. Increases with time
    b. Increases and decreases periodically
    c. Remains constant
    d. Decreases with time

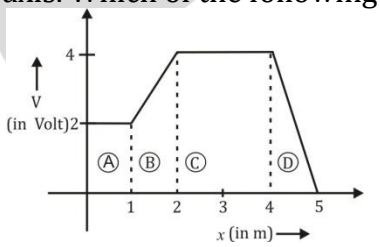
35. Light of certain frequency and intensity incident on a photosensitive material causes photoelectric effect. If both the frequency and intensity are doubled. The photoelectric saturation current becomes.
    a. doubled
    b. unchanged
    c. quadrupled
    d. halved

36. A certain charge 2Q is divided at first into two parts q_1 and q_2. Later, the charges are placed at a certain distance. If the force of interaction between two charges is maximum then $\frac{Q}{q_1} =$ _____.
    a. 2
    b. 0.5
    c. 4
    d. 1

37. A particle of mass m and charge q is placed at rest in uniform electric field E and then released. The kinetic energy attained by the particle after moving a distance y is
    a. qE^2y
    b. q^2Ey
    c. qEy^2
    d. qEy

38. An electric dipole is kept in non-uniform electric field. It generally experiences
    a. A force but not a torque
    b. Neither a force nor a torque
    c. A force and torque
    d. A torque but not a force

39. The figure gives the electric potential V as a function of distance through four regions on x-axis. Which of the following is true for the magnitude of the electric field E in these regions?



    a. $E_A = E_C$ and $E_B < E_D$
    b. $E_A < E_B < E_C < E_D$
    c. $E_A > E_B > E_C > E_D$
    d. $E_B = E_D$ and $E_A < E_C$

40. A system of two charges separated by a certain distance apart stores electrical potential energy. If the distance between them is increased, the potential energy of the system,
    a. Decreases in any case
    b. Remains the same
    c. Increases in any case
    d. May increase or decrease

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41. In a cyclotron, a charged particle
    a. Speeds up between the dees because of the magnetic field.
    b. Slows down with in a dee and speeds up between dees
    c. Undergoes acceleration all the time
    d. Speeds up in dee

42. The number of turns in a coil of Galvanometer is tripled, then
    a. Voltage sensitivity remains constant and current sensitivity increases 3 times
    b. Both voltage and current sensitivity decreases by 33%
    c. Voltage sensitivity increases 3 times and current sensitivity remains constant
    d. Both voltage and current sensitivity remains constant

43. A circular current loop of magnetic moment M is in an arbitrary orientation in an external uniform magnetic field $\vec{B}$. The work done to rotate the loop by 30° about an axis perpendicular to its plane is
    a. $\sqrt{3}\frac{MB}{2}$
    b. Zero
    c. MB
    d. $\frac{MB}{2}$

44. In a permanent magnet at room temperature
    a. The individual molecules have nonzero magnetic moment which are all perfectly aligned.
    b. Domains are all perfectly aligned.
    c. Magnetic moment of each molecule is zero.
    d. Domains are partially aligned.

45. Coersivity of a magnet where the ferromagnet gets completely demagnetized is 3×10^3 Am^{-1}. The minimum current required to be passed in a solenoid having 1000 turns per metre, so that the magnet gets completely demagnetized when placed inside the solenoid is
    a. 60 mA
    b. 6A
    c. 30 mA
    d. 3A

46. An inductor of inductance L and resistor R are joined together in series and connected by a source of frequencyω. The power dissipated in the circuit is
    a. $\frac{V^2R}{R^2+\omega^2L^2}$
    b. $\frac{V^2R}{\sqrt{R^2+\omega^2L^2}}$
    c. $\frac{R^2+\omega^2L^2}{V}$
    d. $\frac{V}{R^2+\omega^2L^2}$

47. An electromagnetic wave is travelling in x-direction with electric field vector given by. $\vec{E}_y = E_0 \sin(kx-\omega t)\hat{j}$. The correct expression for magnetic field vector is
    a. $\vec{B}_z = E_0 C \sin(kx-\omega t)k$
    b. $\vec{B}_z = \frac{E_0}{C}\sin(kx-\omega t)k$

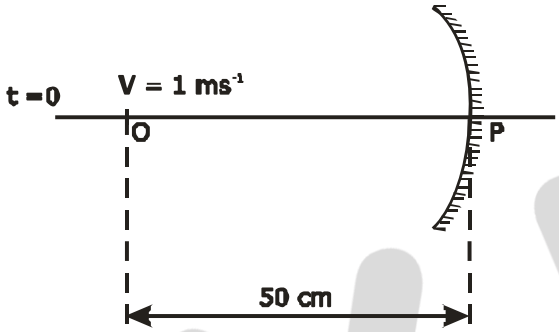
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c. $\vec{B}_y = E_0 C \sin(kx - \omega t)\hat{j}$

d. $\vec{B}_y = \frac{E_0}{C} \sin(kx - \omega t)\hat{j}$

48. The phenomenon involved in the reflection of radio-waves by ionosphere is similar to
    a. Total internal reflection of light in air during a mirage
    b. Scattering of light by air particles
    c. Reflection of light by plane mirror
    d. Dispersion of light by water molecules during the formation of a rainbow

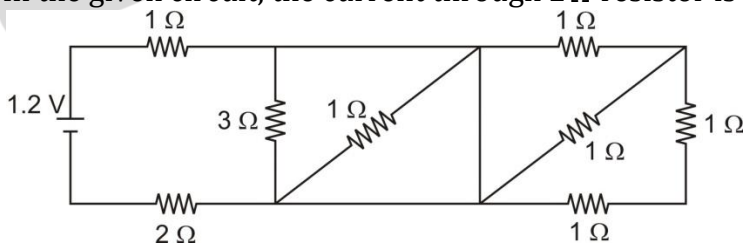
49. A point object is moving uniformly towards the pole of a concave mirror of focal length 25 cm along it's axis as shown below. The speed of the object is $1 ms^{-1}$. At t = 0, the distance of the object from the mirror is 50 cm. The average velocity of the image formed by the mirror between time t = 0 and t = 0.25 s is :



    a. 20 cm s^{-1}
    b. Infinity
    c. 40 cm s^{-1}
    d. Zero

50. A certain prism is found to produce a minimum deviation of 38°. It produces a deviation of 44° when the angle of incidence is either 42° or 62°. What is the angle of incidence when it is undergoing minimum deviation?
    a. 40°
    b. 60°
    c. 30°
    d. 49°

51. In the given circuit, the current through 2 Ω resistor is



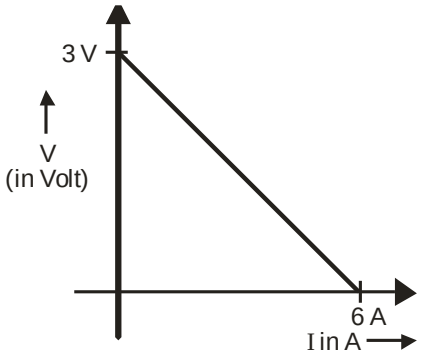
    a. 0.3A
    b. 0.1A
    c. 0.2*A*
    d. 0.4*A*

52. Kirchhoff's junction rule is a reflection of
    a. Conservation of energy
    b. Conservation of charges

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c. Conservation of current density vector
d. Conservation of momentum

53. The variation of terminal potential difference (V) with current flowing through a cell is as shown



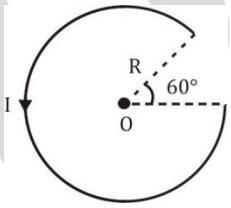
The emf and internal resistance of the cell are

a. 3V, 0.5 Ω
b. 6V, 0.5 Ω
c. 3V, 2 Ω
d. 6V, 2 Ω

54. In a potentiometer experiment, the balancing point with a cell is at a length 240 cm. On shunting the cell with a resistance of 2 Ω, the balancing length becomes 120 cm. The internal resistance of the cell is

a. 2 Ω
b. 0.5 Ω
c. 4 Ω
d. 1 Ω

55. The magnetic field at the centre 'O' in the given figure is



a. $\frac{5}{12}\frac{\mu_0 I}{R}$
b. $\frac{\mu_0 I}{12R}$
c. $\frac{7}{14}\frac{\mu_0 I}{R}$
d. $\frac{3}{10}\frac{\mu_0 I}{R}$

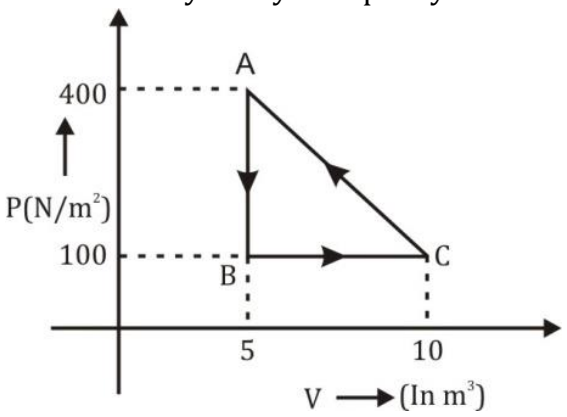
56. An aluminium sphere is dipped into water. Which of the following is true?

a. Buoyancy will be more in water at 0 °C than that in water at 4 °C
b. Buoyancy may be more or less in water at 4 °C depending on the radius of the sphere
c. Buoyancy will be less in water at 0 °C than that in water at 4 °C

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d. Buoyancy in water at 0°C will be same as that in water at 4 °C

57. A thermodynamic system undergoes a cyclic process ABC as shown in the diagram. The work done by the system per cycle is



a. – 1250 J
b. 1250 J
c. 750 J
d. –750 J

58. One mole of O_2 gas is heated at constant pressure starting at 27 °C. How much energy must be added to the gas as heat to double it's volume?

a. 450 R
b. 1050 R
c. Zero
d. 750 R

59. A piston is performing S.H.M. in the vertical direction with a frequency of 0.5 Hz. A block of 10 kg is placed on the piston. The maximum amplitude of the system such that the block remains in contact with the piston is

a. 0.5 m
b. 0.1 m
c. 1 m
d. 1.5 m

60. The equation of a stationary wave is $y = 2 \sin\left(\frac{\pi x}{15}\right) \cos(48\pi t)$. The distance between a node and it's next antinode is

a. 1.5 units
b. 30 units
c. 7.5 units
d. 22.5 units

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ANSWER KEYS

1. (c)	2. (b)	3. (d)	4. (d)	5. (c)	6. (c)	7. (c)	8. (a)	9. (c)	10. (b)
11. (d)	12. (a)	13. (d)	14. (c)	15. (b)	16. (c)	17. (d)	18. (a)	19. (a)	20. (d)
21. (c)	22. (b)	23. (c)	24. (a)	25. (a)	26. (a)	27. (d)	28. (c)	29. (c)	30. (c)
31. (d)	32. (a)	33. (d)	34. (c)	35. (a)	36. (d)	37. (d)	38. (c)	39. (a)	40. (d)
41. (c)	42. (a)	43. (b)	44. (d)	45. (d)	46. (a)	47. (b)	48. (a)	49. (b)	50. (d)
51. (d)	52. (b)	53. (a)	54. (a)	55. (a)	56. (c)	57. (d)	58. (b)	59. (c)	60. (c)

BYJU'S

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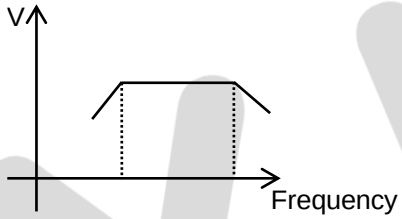
KCET-2019 (Physics)

Solution

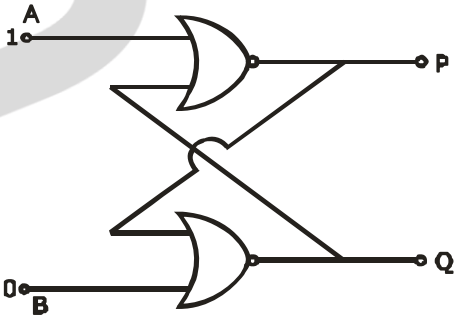
1. (c)
Activity is given by slope of the graph, higher activity implies shorter mean life.
Slope of A is greatest $\Rightarrow$ higher activity $\Rightarrow$ shorter mean life

2. (b)
As temperature increases, the number density of charge carriers increases according to the equation
$n = CT^{3/2}e^{\left(-\frac{E}{2KT}\right)}$
Thus as temperature increases, the number density of charge carriers increases resulting in increased conductivity of semiconductors.

3. (d)
The frequency response curve of the transistor amplifier shows that the voltage gain of a transistor amplifier is constant at mid frequency range only. It is low at high and low frequencies.



4. (d)
For both NOR gates
Output: $Y = \overline{A + B}$
For gate 1; A = 1 $\therefore$ P = 0
For gate 2; B = 0 $\therefore$ Q =1



5. (c)
$C = f\lambda$
$\lambda = c/f = \frac{3\times10^8}{5\times10^6} = \frac{3}{5}\times10^2 = 60\text{m}$

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$\therefore$ Antenna length = $\frac{\lambda}{4} = \frac{60}{4} = 15$ m

6. (c)

$T = 2\pi\sqrt{\frac{I}{MB}}$

$2 = 2\pi\sqrt{\frac{8\times10^{-6}}{5\times10^{-2}\times B}}$

$\Rightarrow B = 1.6 \times 10^{-4}$ T

7. (c)

Magnetic field inside a toroid

$B = \mu_0 NI$

Magnetic energy density inside the toroid

$= \frac{B^2}{2\mu_0}$

$= \frac{\mu_0^2 N^2 I^2}{2\mu_0}$

$= \frac{\mu_0 N^2 I^2}{2}$

$= \frac{4\pi\times10^{-7}\times(500)^2\times2^2}{2}$

$= 0.628$ J/m^3

8. (a)

Induced emf.: $\varepsilon = \ell.\ (\vec{V}\times\vec{B})$

$= (BV\sin\theta)\ \ell$

$= BV\ \ell$

$\therefore$ Induced current: $I = \frac{\varepsilon}{R}$

$\therefore$ If the straight wire is replaced by a semicircular wire resistance remain the same, therefore current remains same

9. (c)

Time period: $T = \frac{1}{f} = \frac{1}{50}$ sec

Time taken by current to reach peak value in Ac = $\frac{T}{4}$

$\Rightarrow \frac{T}{4} = \frac{1}{50\times4} = \frac{1}{200}$

$= 5 \times 10^{-3}$ S

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10. (b)

As evident from question

$V_L = V_C$

$\Rightarrow$ Circuit is purely resistive

$\therefore$ V = 220 V

Also; $I = \frac{V}{R} = \frac{220}{100}$

$\Rightarrow$ I = 2.2 A

11. (d)

Electric field: $E = \frac{V}{d}$

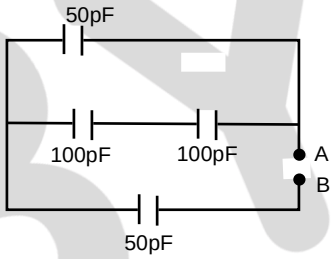
$V \rightarrow$ Potential difference

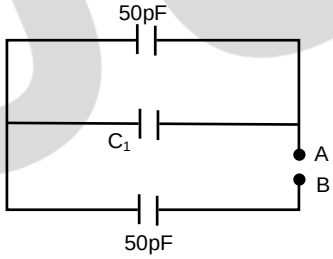
$d \rightarrow$ separation between plates

$\therefore$ $E = \frac{30-(-10)}{2\times10^{-2}}$

$\therefore$ E = 2000 V/m

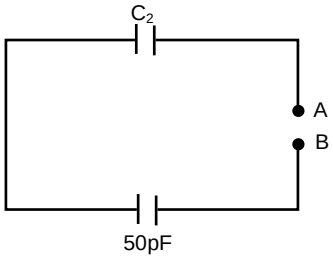
12. (a)





$\Rightarrow$ $C_1 = \frac{100\times100}{100+100}$ = 50 pF

$\Rightarrow$



$C_2 = 50 + C_1$

= 50 + 50

= 100pF

Now; C_2 & 50pF are in series across AB

$\therefore C_{AB} = \frac{50\times100}{50+100} = \frac{5000}{150}$

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$\therefore \quad C_{AB} = \frac{100}{3}\text{pF}$

13. (d)

Let the common potential attained by the two capacitor be "V_c"

$\therefore \quad V_c = \frac{\text{Total charge}}{\text{Total capacitance}} = \frac{Q+0}{C+2C}$

$\Rightarrow \quad V_C = \frac{Q}{3C}$

$\therefore$ Final charge on capacitors

$Q_1 = CV_c = \frac{CQ}{3C} = \frac{Q}{3}$

$Q_2 = 2CV_c = \frac{2Q}{3}$

14. (c)

Current: $I = neAV_d$

The large value of I is due to number density of free electron in a conductor (n),

Which is of the order of n= 10^{28} per m^3

15. (b)

Resistance of wire: $R = \frac{\rho L}{A}$

$\rho \rightarrow$ resistivity of wire

Area: $A = \frac{\text{Volume}}{\text{Length}} = \frac{V}{L}$, Volume = $\frac{\text{Mass}}{\text{Density}} = \frac{m}{d}$

$\Rightarrow \quad R = \rho d \frac{L^2}{m}$

$\Rightarrow \quad R \propto \frac{L^2}{m}$

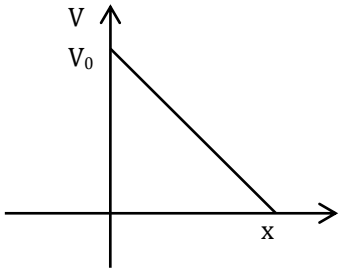
$\therefore \quad R_1 : R_2 : R_3 = \frac{L_1^2}{m_1} : \frac{L_2^2}{m_2} : \frac{L_3^2}{m_3} = \frac{5^2}{1} : \frac{3^2}{3} : \frac{1^2}{5}$

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$\therefore \quad R_1 : R_2 : R_3 = 125 : 15 : 1$

16. (c)
By principle of Homogeneity; Addition & subtraction of physical quantities having different dimensions is not possible

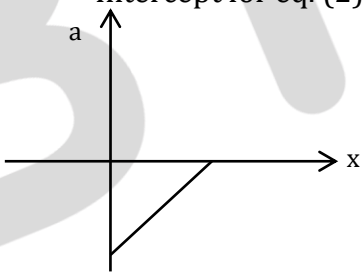
17. (d)
The equation for the given graph can be written as.



$V = -mx + v_0$ ..... (1)
$m \rightarrow$ slope of graph which is –ve
$V_0 \rightarrow$ Intercept of line which is +V and a constant

$\therefore \quad a = \frac{dv}{dt} = -m\frac{dx}{dt} + \frac{d}{dt}V_0$

$\Rightarrow \quad a = -mV + 0$
But from eq (1)
$a = -m(-mx + V_0)$
$\Rightarrow \quad a = +m^2x - mV_0$ .... (2)
$\therefore$ Slope for given eq. (2): +Ve
Intercept for eq. (2):–Ve



18. (a)

Given; $y = x - \frac{2x^2}{5}$

Comparing with standard equation of trajectory

i.e. $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\therefore \quad \tan\theta = 1$

$\Rightarrow \quad \theta = 45^o$

Also $\frac{g}{2u^2\cos^2\theta} = \frac{2}{5}$

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$\Rightarrow \quad \frac{g}{2u^2 \times \frac{1}{2}} = \frac{2}{5}$

$\Rightarrow \quad u = 5\ m/s$

19. (a)

$\vec{F} = -3\hat{i} + 4\hat{j}$

$\Rightarrow \quad \vec{a} = \frac{\vec{F}}{m} = = \frac{-3}{5}\hat{i} + \frac{4}{5}\hat{j}$

$\Rightarrow \quad a_x = \frac{-3}{5}$

At $t = 0;\ V_0 = 6\hat{i} - 12\hat{j}$

$\therefore \quad V_{ox\ at\ t=0} = 6$

For body to have velocity only along y-axis, velocity along x-axis should be zero

$\therefore \quad V_x = V_{ox} + a_x t$

$0 = 6 - \left(\frac{3}{5}\right)t$

$\Rightarrow \quad t = 10\ sec$

20. (d)

Linear momentum always remains conserved for any type of collision

21. (c)

For head on collision impact parameter is zero as it retraces its path

22. (b)

Time period of revolution; $T = \frac{2\pi r}{V}$

$\therefore$ frequency; $f = \frac{v}{2\pi r}$

Also; $V \propto \frac{1}{n}$

And $r \propto n^2$

(By Bohr's model)

$\therefore \quad f \propto \frac{1}{n^3}$

23. (c)

By absorbing 10.2 eV electron goes to 2nd orbit, ie n = 2

$\therefore$ Angular momentum increased by

$L_2 - L_1 = \frac{2h - h}{2\pi}$

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$= \frac{h}{2\pi} = \frac{6.6\times10^{-34}}{2\times3.14} = 1.05 \times 10^{-34}$ JS

24. (a)
$_{90}Th^{232} \longrightarrow {}_{82}Pb^{208}$
Hence ; Mass number changes by 24 this implies 6 α particles are emitted as 1 α particle emitted decreases mass number by 4, then proton number should decrease by 12, but change in proton number is by 8 only, this implies 4β particles have been emitted.

25. (a)
As separation is less than 10nm, therefore $F_e >> F_n$ as nuclear force is short range force.

26. (a)
Initially both A & B are at rest, hence initial momentum is zero, Also no external force is acting on the two particles as they are moving under mutual attraction so final momentum should also be zero, by conservation of linear momentum.
$\therefore$ $V_{com} = 0$, as mass cannot be zero

27. (d)
Angular momentum = mvr
Here r is perpendicular distance of particle from reference point i.e. O, so as m, v, r are constant for position A & B.
$\therefore$ Angular momentum remains constant.

28. (c)
We known that escape velocity
$V_e = \sqrt{2}\ v_0$
$V_0 \rightarrow$ orbital velocity
When satellite is in orbit; $K = \frac{1}{2}\ mv_0^2$
$\therefore$ Kinetic energy required to escape; $K' = \frac{1}{2}\ mv_e^2 = mv_0^2 = 2K$
$\therefore$ Extra Kinetic energy = 2K – K = K

29. (c)
Let L be the length and r be the radius of wire.
$\therefore$ Volume: $V = \pi r^2 L$
Differentiating both sides
$\Delta V = \pi (2r\Delta r)L + \pi r^2 \Delta L$
As volume remains unchanged $\therefore \Delta L = 0$
$\therefore$ $O = 2\pi rL\Delta r + \pi r^2 \Delta L$
$\therefore$ $\frac{\Delta r / r}{\Delta L / L} = \frac{-1}{2}$

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Poisson's ratio = $\frac{\text{Lateral strain}}{\text{Longitudinal strain}}$

$= -\frac{\Delta r / r}{\Delta L / L} = \frac{1}{2}$ 0.5.

30. (c)

$V_x = \sqrt{2gh}$ $(h = 2 \times 10^{-2})$

$= \sqrt{2 \times 10 \times 2 \times 10^{-2}}$

$= \sqrt{\frac{4}{10}} = \sqrt{\frac{2}{5}}$

$H = 8 \times 10^{-2} m$

$g = 10 m/s^2$

$\therefore t = \sqrt{\frac{2H}{g}}$

$t = \sqrt{\frac{2 \times 8 \times 10^{-2}}{10}}$

$\boxed{t = \frac{2}{5\sqrt{10}} sec}$

$\therefore$ Maximum horizontal distance; $x = v \times t = \sqrt{\frac{2}{5}} \times \frac{2}{5\sqrt{10}} = \frac{2\sqrt{2}}{5\sqrt{50}} = \frac{2}{25} m = 8$ cm

31. (d)

By Brewster's law; $n = \tan\theta_p$

Also; $n = \frac{\sin i}{\sin r}$

But $\frac{\sin i}{\sin r} = \frac{1}{\text{Slope}}$

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$n = \frac{\sin i}{\sin r} = \frac{1}{\tan 30^o} = \sqrt{3}$

$n = \tan\theta_p = \sqrt{3}$

$\therefore \quad \theta_p = \tan^{-1}\sqrt{3}$

$\therefore \quad \theta_p = 60^o$

32. (a)

We know that

$$\Delta x = \frac{\lambda}{2\pi}\Delta\phi$$

$\therefore \quad \frac{\lambda}{3} = \frac{\lambda}{2\pi}\Delta\phi$

$\therefore$ Phase difference; $\phi = \frac{2\pi}{3}$

$\therefore$ Intensity; $I = I_0 \cos^2\phi$

$\therefore \quad I = K\cos^2\left(\frac{2\pi}{3}\right)$

$$I = K\left(\frac{-1}{2}\right)^2$$

$$I = \frac{K}{4}$$

33. (d)

Doppler shift: $\frac{\Delta\lambda}{\lambda} = \frac{V}{C}$

$\therefore \quad V = \frac{\Delta\lambda}{\lambda}.C$

$\therefore \quad V = \frac{0.1}{6000} \times 3 \times 10^8$

$\therefore \quad V = 5\text{km/s}$

34. (c)

Magnetic force on moving electron:

$$F = -e\ (V_0\hat{i} \times B_0\hat{j})$$

$\Rightarrow \quad F = -eV_0 B_0 \hat{k}$

So, Force is perpendicular to both $\vec{V}$ & $\vec{B}$ so the magnitude of V will not change it means momentum will not change.

Now; by De – Broglie

$\lambda = \frac{h}{mv} = \frac{h}{p}$

$\therefore$ As momentum remains constant

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$\Rightarrow$De Broglie wavelength remains constant

35. (a)
We know that photo electric current is directly proportional to the intensity of light, given that incident frequency is greater than threshold frequency.
$\therefore$ When intensity is doubled.
$\Rightarrow$Photo electric saturation current doubles

36. (d)
Force of interaction between two charges q_1 & q_2 will be maximum when

$q_1 = q_2 = \frac{2Q}{2} = Q$

$\therefore \quad \frac{Q}{q_1} = \frac{Q}{Q} = 1$

37. (d)
Velocity gained after moving a distance y:

$V^2 = u^2 + 2ay$

Here; $u = 0$,

$F = qE$

$ma = qE$

$\Rightarrow \quad a = \frac{qE}{m}$

$\therefore \quad V^2 = 0 + 2\left(\frac{qE}{m}\right)y$

$\therefore$ Kinetic energy

$E = \frac{1}{2} mv^2$

$E = \frac{1}{2} \times m \times \left(2 \times \frac{qE}{m} \times y\right)$

$E = qEy$

38. (c)
An electric dipole in non-uniform electric field generally experiences a force and a torque.

39. (a)

In region A, $V \rightarrow$ constant $\therefore$ $\boxed{E_A = 0}$ as $E = \frac{dV}{dx}$

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In region B, $E_B = \frac{4-2}{2} = 1$ v/m $\Rightarrow$ $E_B = 1V/m$

In region C, $V \rightarrow$ constant $\Rightarrow$ $E_C = 0$

In region D, $E_D = \frac{4-2}{1} = 2$v/m $\Rightarrow$ $E_D = 2$ V/M

$\therefore$ $E_A = E_C$ & $E_D > E_B$

40. (d)
Potential energy of system of 2 charges

$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$

$r \rightarrow$ separation between 2 charges q_1 & q_2
If both charges are of same type, then on increasing the separation, potential energy decreases.
If the charges are of opposite type, then on increasing separation, potential energy increases.

41. (c)
In a cyclotron a charged particle experiences electric force between the dees and a magnetic force inside the dees while circulating, hence it always experiences a force and hence acceleration.

42. (a)
Current sensitivity: $S_I = \frac{\phi}{I} = \frac{NBA}{K}$

Voltage sensitivity: $S_V = \frac{\phi}{V} = \frac{NIBA}{KV} = \frac{NBA}{KR}$

When we triple "N" then R is also tripled

Hence; $\frac{\phi}{I} \rightarrow$ becomes 3 times

But $\frac{\phi}{V} \rightarrow$ Remain unchanged

43. (b)
Even when the coil is rotated, about an axis perpendicular to its plane, the potential energy does not change.
Hence, work done is zero

44. (d)
Even in case of permanent magnets all the domain are not perfectly aligned due to thermal agitation

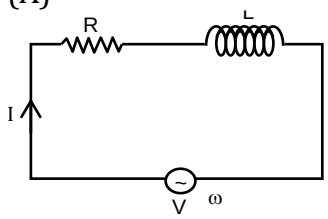
45. (d)
Coercivity refers to the intensity of magnetic field (B) where the magnets get demagnetized
For solenoid; $B = \mu_0 n I$
$n \rightarrow$ No of turns per unit length

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$I \rightarrow$ current through each turn
$\mu_0 \rightarrow$ Absolute permeability
$B \rightarrow$ magnetic field in solenoid
Also $\mu_0 H = \mu_0 nI$
$H \rightarrow$ intensity of magnetic field
$\therefore \quad I = H/n$

$$I = \frac{3\times10^3}{\frac{100}{0.1}} = 3A$$

46. (A)



$$Z = \sqrt{R^2 + X_L^2}$$

$$Z = \sqrt{R^2 + (\omega L)^2}$$

$\therefore \quad P = I^2R$

$$I = \frac{V}{\sqrt{R^2 + \omega^2 L^2}}$$

$$\therefore \quad P = \frac{V}{(R^2 + \omega^2 L^2)}$$

47. (B)

$$\vec{E_y} = E_0 \sin(Kx - \omega t)\ \hat{j}$$

$$C = \frac{E_0}{B_0}$$

$$\Rightarrow B_0 = \frac{E_0}{C}$$

$$\therefore \quad \vec{B_z} = \frac{E_0}{C} \sin(Kx - \omega t)\ \hat{k}$$

48. (A)
The phenomenon involved in the reflection of radio-waves by ionosphere is similar to total internal reflection of light in air during a mirage

49. (B)
Focal length = 2.5cm
At t = 0, object is at a distance of x_i =50cm

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At t = 0.25 object is at a distance of x_f = speed × time

=0.1 × 0.25

= 0.25 m

= 25 cm

$\therefore$ At t = 0.25 image will beat infinity

$\therefore$ $V_{avg} = \frac{\Delta x}{\Delta t} = \infty$

50. (d)

$\delta_{min} = 38^o$

$i = \frac{A+\delta}{2}$

$\delta \rightarrow$ deviation

$i \rightarrow$ angle of incidence

$A \rightarrow$ Angle of prism

$\delta = (r_1 + r_2) - A$

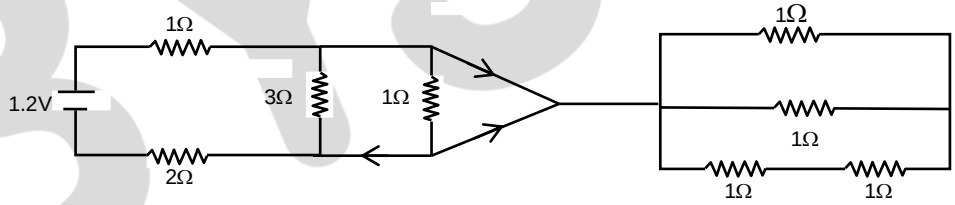
$-A = 44 - 104$

$\therefore$ $A = 60^o$

$\Rightarrow$ $i = \frac{60+38}{2} = 49$

51. (d)

The circuit can be redrawn as.



$\therefore$ $R_{eq} = 1 + 2 = 3\Omega$

$I = \frac{V}{R_{eq}} = \frac{1.2}{3} = 0.4$ A

52. (b)

Kirchhoff's junction rule is based on conservation of charges.

53. (a)

From the given graph, we can form the following equation

V = E – Ir

When I = 0, $\Rightarrow$ E = V = 3V

When V = 0 $\Rightarrow$ I = 6A

$\therefore$ $r = \frac{E}{I} = \frac{3}{6} = 0.5\ \Omega$

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54. (a)

Internal resistance of a cell is given by:

$r = R\left(\frac{\ell_1}{\ell_2} - 1\right)$

$r = 2\left(\frac{240}{120} - 1\right)$

$r = 2 \times 1$

$r = 2\Omega$

55. (a)

Here $\theta = 300°$ as can be seen from given diagram

$\therefore$ Number of turns; $n = \frac{300}{360} = \frac{5}{6}$

$$B = \mu_0 \frac{nI}{2R} = \mu_0 \times \frac{5}{6} \times \frac{I}{2R}$$

$$\therefore \quad B = \frac{5\mu_0 I}{12R}$$

56. (c)

Buoyancy at 0°C; $F_B = V\rho_0 g$

Buoyancy at 4°C; $F_B' = V\rho_4 g$

$$\frac{F_B}{F_B'} = \frac{\rho_0}{\rho_4}$$

As we know ρ at 4°C is more than ρ at 0°C

$\therefore \quad \rho_4 > \rho_0$

$\Rightarrow \quad F_B < F_B'$

57. (d)

Work done = $P\Delta V$

Also; work done = area of ΔABC

$W = \frac{1}{2}(10-5) \times (100 - 400)$

$W = -750$ J

58. (b)

If at constant pressure, volume is doubled then temperature also gets doubled.

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$T_1 = 300K, \quad T_2 = 600\ K$

$\therefore \quad \Delta T = 600 - 300 = 300\ K$

$\therefore \quad Q = nC_p\Delta T = 1 \times \frac{7}{2} R \times 300$

$= 1050\ R$

59. (c)

$f = 0.5\ Hz$

$\omega = 2\pi f$

$= 2 \times \pi \times \frac{1}{2}$

$= \pi$

For block to remain in contact with the piston, at amplitude position of SHM

Weight of block = force due to oscillation

$mg = ma$

$mg = m(\omega^2 A)$

$A = \frac{g}{\omega^2} = \frac{10}{\pi^2}$

$\therefore \quad A \cong 1m$

60. (c)

$y = 2 \sin\left(\frac{\pi x}{15}\right) \cos(48\pi t)$

$\frac{2\pi}{\lambda} = \frac{\pi}{15}$

$\Rightarrow \quad \lambda = 30$

$\therefore$ Distance between node & next antinodes

$\Rightarrow \quad \frac{\lambda}{4} = \frac{30}{4} = 7.5$

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